CORE S115 Fall 2026: Lab 6

One predictor

Generate some data

```
x = c(1:100)
y1 = 10 + x*0.5 + rnorm(100, 0, 5)
y2 = 20 + x*-2 + rnorm(100, 0, 5)
y3 = 40 + rnorm(100, 0, 5)

dat= data.frame(x=x,
                y1=y1,
                y2=y2,
                y3=y3)
```

Comprehension check In the simulated data, what do you think is the relationship between:

- x and y1
- x and y2
- x and y3

Fit regression model

Here is code that fits regression models for each of the three y values.

```
fit1 = lm(y1 ~ x, data = dat)
summary(fit1)
##
## Call:
## lm(formula = y1 ~ x, data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.8948  -2.7806  -0.3334   2.9439  12.9179
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   9.3512     0.9770   9.571 1.03e-15 ***
## x             0.5168     0.0168  30.767  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 4.848 on 98 degrees of freedom
## Multiple R-squared:  0.9062, Adjusted R-squared:  0.9052
## F-statistic: 946.6 on 1 and 98 DF,  p-value: < 2.2e-16
```

```
fit2 = lm(y2 ~ x, data = dat)
summary(fit2)
##
## Call:
## lm(formula = y2 ~ x, data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.0231  -3.1610   0.1685   3.4111  10.3052
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 21.65422    0.99937   21.67   <2e-16 ***
## x           -2.01904    0.01718 -117.52   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.959 on 98 degrees of freedom
## Multiple R-squared:  0.993,  Adjusted R-squared:  0.9929
## F-statistic: 1.381e+04 on 1 and 98 DF,  p-value: < 2.2e-16
```

```
fit3 = lm(y3 ~ x, data = dat)
summary(fit3)
##
## Call:
## lm(formula = y3 ~ x, data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.1157  -3.9519   0.1582   3.5245  12.3916
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 40.353700   1.104514  36.535   <2e-16 ***
## x            0.004386   0.018988   0.231    0.818
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.481 on 98 degrees of freedom
## Multiple R-squared:  0.0005442,  Adjusted R-squared:  -0.009654
## F-statistic: 0.05336 on 1 and 98 DF,  p-value: 0.8178
```

Comprehension check What do you notice about the coefficient values and the p-values? Does this make sense? Why is the intercept different in each case?

Two predictors

Here is code that will generate y values that are a combination of two predictors, x1 and x2.

```
x1 = c(1:100)
x2 = rep(c(0, 1), times=50)

y1 = 50 + x1*0.5 + x2*-20 + rnorm(100, 0, 5)
y2 = 50 + x1*-2 + x2*0.01 + rnorm(100, 0, 10)

dat2= data.frame(x1=x1,
                 x2=x2,
                 y1=y1,
                 y2=y2)
```

Comprehension check In the simulated data, what do you think is the relationship between:

- x1, x2 and y1
- x1, x2 and y2

Fit regression model

Here is code that fits regression models with two predictors.

```
fit4 = lm(y1 ~ x1 + x2, data=dat2)
summary(fit4)
##
## Call:
## lm(formula = y1 ~ x1 + x2, data = dat2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.8245  -2.8384   0.2973   2.6232  12.0694
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  48.34831    1.08087   44.73   <2e-16 ***
## x1            0.50592    0.01675   30.21   <2e-16 ***
## x2          -18.15131    0.96678  -18.77   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.833 on 97 degrees of freedom
## Multiple R-squared:  0.9278, Adjusted R-squared:  0.9263
## F-statistic:   623 on 2 and 97 DF,  p-value: < 2.2e-16
```

```
fit5 = lm(y2 ~ x1 + x2, data=dat2)
summary(fit5)
##
## Call:
```

```
## lm(formula = y2 ~ x1 + x2, data = dat2)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -26.859  -6.888  -0.391   8.525  20.314
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 49.09394    2.25132  21.807   <2e-16 ***
## x1          -1.95872    0.03488 -56.156   <2e-16 ***
## x2          -0.21195    2.01370  -0.105    0.916
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.07 on 97 degrees of freedom
## Multiple R-squared:  0.9702, Adjusted R-squared:  0.9696
## F-statistic:  1577 on 2 and 97 DF,  p-value: < 2.2e-16
```

Comprehension check What do you notice about the coefficient values and the p-values? Does this make sense? Specifically consider the coefficient and pvalue for x2 in fit5.

Interactions

Here is code that will generate y values that are a combination of two predictors, x1 and x2, and where the two predictors interact.

```
x1 = c(1:100)
x2 = rep(c(0, 1), times=50)

y1 = 50 + x1*0.5 + x2*-20 + x1*x2*1 + rnorm(100, 0, 5)
y2 = 50 + x1*-2 + x2*10 + x1*x2*-0.01 + rnorm(100, 0, 5)

dat2= data.frame(x1=x1,
                 x2=x2,
                 y1=y1,
                 y2=y2)
```

```
fit6 = lm(y1 ~ x1 + x2 + x1:x2, data=dat2)
summary(fit6)
##
## Call:
## lm(formula = y1 ~ x1 + x2 + x1:x2, data = dat2)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -11.758  -3.468   0.749   2.725   9.040
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  50.49972    1.19228   42.35   <2e-16 ***
```

```
## x1            0.46765    0.02065   22.64   <2e-16 ***
## x2          -21.26635    1.69887  -12.52   <2e-16 ***
## x1:x2         1.04128    0.02921   35.65   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.215 on 96 degrees of freedom
## Multiple R-squared:  0.9871, Adjusted R-squared:  0.9867
## F-statistic:  2440 on 3 and 96 DF,  p-value: < 2.2e-16
```

```
fit7 = lm(y2 ~ x1 + x2 + x1:x2, data=dat2)
summary(fit7)
##
## Call:
## lm(formula = y2 ~ x1 + x2 + x1:x2, data = dat2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.5284  -3.1158  -0.1929   2.8831  12.0129
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 50.224880   1.425458  35.234  < 2e-16 ***
## x1          -1.996554   0.024691 -80.862  < 2e-16 ***
## x2          10.560586   2.031117   5.199 1.13e-06 ***
## x1:x2       -0.008104   0.034918  -0.232    0.817
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.039 on 96 degrees of freedom
## Multiple R-squared:  0.9928, Adjusted R-squared:  0.9926
## F-statistic:  4399 on 3 and 96 DF,  p-value: < 2.2e-16
```

Factors

Categorical variable with two categories

When data is categorical R automatically assigns one of the predictors as the baseline – specifically the predictor which is alphabetically first. Consider the following case.

```
x = rep(c('Instrumental', 'Noise'), times=50)
x_num = ifelse(x=='Noise', 0, 1)

y = 65 + x_num*10 + rnorm(100, 0, 5)


dat3 = data.frame(x=x,
                  y=y)
```

```
fit8 <- lm(y~x, data=dat3)
summary(fit8)
##
## Call:
## lm(formula = y ~ x, data = dat3)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.9567 -3.5141 -0.3384  3.2237 12.9117
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  75.3978     0.6240   120.8   <2e-16 ***
## xNoise      -11.2978     0.8825   -12.8   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.412 on 98 degrees of freedom
## Multiple R-squared:  0.6258, Adjusted R-squared:  0.622
## F-statistic: 163.9 on 1 and 98 DF,  p-value: < 2.2e-16
```

Comprehension check Why is the coefficient negative? What does this tell you about what is being considered a "baseline"? Does this make sense?

Categorical variable with more than two categories

And if the variable has more than two categories (or "levels") then it assigns the first alphabetical as the "baseline" and computes a coefficient comparing the baseline to each of the others.

```
x = rep(c('Pop', 'Instrumental', 'Noise'), times=50)
x_num = ifelse(x=='Noise', 0, 1)

y = 65 + x_num*10 + rnorm(150, 0, 5)


dat4 = data.frame(x=x,
                  y=y)
```

```
fit9 <- lm(y~x, data=dat4)
summary(fit9)
##
## Call:
## lm(formula = y ~ x, data = dat4)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.3359  -3.0636   0.3327   2.9678   9.6756
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)  74.4844     0.6442 115.628   <2e-16 ***
## xNoise       -9.5831     0.9110 -10.519   <2e-16 ***
## xPop          1.0151     0.9110   1.114    0.267
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.555 on 147 degrees of freedom
## Multiple R-squared:  0.5286, Adjusted R-squared:  0.5222
## F-statistic: 82.41 on 2 and 147 DF,  p-value: < 2.2e-16
```

Comprehension check Why is one of the coefficients negative and one positive? What does this tell you about what is being considered the "baseline"? Does this baseline make sense?

Changing what is assigned as the baseline

The following code explicitly changes the ordering of the categories, and therefore what is considered baseline

```
dat5 = dat4 %>%
  mutate(x = factor(x, levels=c('Noise', 'Instrumental', 'Pop')))
```

```
fit10 <- lm(y~x, data=dat5)
summary(fit10)
##
## Call:
## lm(formula = y ~ x, data = dat5)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.3359  -3.0636   0.3327   2.9678   9.6756
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    64.9013     0.6442  100.75   <2e-16 ***
## xInstrumental   9.5831     0.9110   10.52   <2e-16 ***
## xPop           10.5982     0.9110   11.63   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.555 on 147 degrees of freedom
## Multiple R-squared:  0.5286, Adjusted R-squared:  0.5222
## F-statistic: 82.41 on 2 and 147 DF,  p-value: < 2.2e-16
```

Comprehension check What could you do if you wanted to set Pop as the baseline?

Extra comprehension check (its ok if you don't fully get this!) You will notice that only the intercept in `fit8` matches the number we were adding. The other two intercepts were off by about 10. Why do you think this is? *Hint: look at* ***x_num*** *and what role it plays in the data generation!*